

The Complex Arctangent

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Overview

We study the complex function $\arctan z$ via

- its logarithmic representation,
- its integral definition,
- partial fractions,
- and power series expansions about different centers.

From $\tan w = z$ to a Logarithm

Let $w = \arctan z$, so $z = \tan w$. Using the exponential form,

$$\tan w = \frac{e^{iw} - e^{-iw}}{i(e^{iw} + e^{-iw})}.$$

Setting $u = e^{2iw}$ gives

$$z = \frac{u - 1}{i(u + 1)}, \quad e^{2iw} = \frac{1 + iz}{1 - iz}.$$

Closed Form

Taking logarithms,

$$2iw = \log\left(\frac{1+iz}{1-iz}\right),$$

hence

$$\boxed{\arctan z = \frac{1}{2i} \log\left(\frac{i+z}{i-z}\right)}.$$

Branch points occur at $z = \pm i$.

Integral Definition

Analytically,

$$\arctan z = \int_0^z \frac{1}{1+t^2} dt, \quad t \neq \pm i.$$

We now analyze the integrand.

Partial Fractions

Factor

$$1 + t^2 = (t - i)(t + i).$$

Solve

$$\frac{1}{1 + t^2} = \frac{A}{t - i} + \frac{B}{t + i}.$$

This yields

$$\boxed{\frac{1}{1 + t^2} = \frac{1}{2i} \left(\frac{1}{t - i} - \frac{1}{t + i} \right)}.$$

Recovering the Logarithmic Formula

Insert the partial fractions:

$$\arctan z = \frac{1}{2i} \int_0^z \left(\frac{1}{t-i} - \frac{1}{t+i} \right) dt.$$

Integrating,

$$\arctan z = \frac{1}{2i} [\log(t-i) - \log(t+i)]_0^z.$$

Up to branch constants,

$$\arctan z = \frac{1}{2i} \log \left(\frac{i+z}{i-z} \right).$$

Maclaurin Series via Partial Fractions

For $|t| < 1$,

$$\frac{1}{t \pm i} = \frac{1}{\pm i} \frac{1}{1 \mp t/i} = \frac{1}{\pm i} \sum_{n=0}^{\infty} \left(\pm \frac{t}{i} \right)^n.$$

Substitute into the partial fraction formula.

Simplification

Combining the two series,

$$\frac{1}{1+t^2} = \sum_{n=0}^{\infty} (-1)^n t^{2n}, \quad |t| < 1.$$

Integrating term-by-term,

$$\arctan z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{2n+1}.$$

Radius of Convergence

The nearest singularities occur at

$$z = \pm i,$$

so the Maclaurin series has radius of convergence

$$R = 1.$$

Expansion About $z = 1$

Let

$$z = 1 + w, \quad |w| \ll 1.$$

Then

$$\arctan z = \arctan 1 + \int_0^w \frac{1}{1 + (1 + s)^2} ds, \quad \arctan 1 = \frac{\pi}{4}.$$

Using Partial Fractions Again

Write

$$\frac{1}{1 + (1 + s)^2} = \frac{1}{(s + 1 - i)(s + 1 + i)}.$$

Apply partial fractions:

$$\frac{1}{1 + (1 + s)^2} = \frac{1}{2i} \left(\frac{1}{s + 1 - i} - \frac{1}{s + 1 + i} \right).$$

Series Expansion

Expand each term as a geometric series about $s = 0$:

$$\frac{1}{s + 1 \pm i} = \frac{1}{1 \pm i} \sum_{n=0}^{\infty} \left(-\frac{s}{1 \pm i} \right)^n.$$

Integrate term-by-term to obtain a power series in $w = z - 1$.

Resulting Taylor Series

Collecting terms,

$$\arctan z = \frac{\pi}{4} + \frac{1}{2}(z - 1) - \frac{1}{4}(z - 1)^2 + \frac{1}{12}(z - 1)^3 + \dots$$

The radius of convergence is the distance from $z = 1$ to $z = i$.

Final Remarks

- Partial fractions unify all representations of $\arctan z$
- Singularities at $\pm i$ control convergence and branching
- Different centers give genuinely different local descriptions